From Model to Plant Floor

Industry-adoption roadmap

Muhammet Yalkapov, Sulaymon Sadullo, Rama Abdallah, Khadeja Al Khazraji
Abu Dhabi University · Team HYDRA · Registration CMAT-00107
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What is actually being offered

Not a plant retrofit. The adoptable artefact is a **decision-support model** that answers one question an operator cannot currently answer:

> *If I flex this plant to absorb more solar, what does it cost me in membrane life — and is that trade worth taking?*

The model is open, runs on a laptop in minutes, and requires no proprietary data to produce a first answer. Its value to an operator rises sharply once their own trend data is loaded, which is the basis of the staged plan below.

Who adopts it, and why they would

Stakeholder	Decision they face	What this gives them
Plant operators (e.g. IWP concession holders)	Whether to accept a flexible-dispatch clause in a PPA or offtake agreement	A quantified membrane-life cost for flexibility, and a defensible basis for declining it
Utility / offtaker (EWEC and equivalents)	How much desalination load can be counted as dispatchable	Evidence that it should not be counted as dispatchable at present membrane capability
Developers and EPCs	How much PV and RO oversize to build	A design-space map showing where the cost optimum actually sits
Membrane manufacturers	Where to direct product development	A hard target: $N^{*}_{\mathrm{tol}} \approx 4201$ cycles is the number to beat
Regulators and planners	Instrument design for decarbonisation	The carbon-price schedule, and a warning on tariff sequencing

The adoption barrier, stated honestly

The model's central parameter — irreversible permeability loss per start/stop cycle — is unmeasured. An operator cannot act on our specific numbers; they can act on our *method* and on the asymmetry we identify (upside at most 0.2%, downside up to 47%). Everything below is sequenced so that the measurement happens early and cheaply, because until it does the model informs judgement rather than replacing it.

Staged roadmap

Stage 0 — Validate against one real plant (0–6 months)

Objective. Replace assumed fouling coefficients with fitted ones.

- Obtain, for one train over one year: normalised permeate flow or specific flux, normalised differential pressure, cleaning event log, membrane replacement dates, and the running/idle state history.
- All acceptable in normalised, indexed or date-shifted form; the analysis needs trend shapes, not absolute performance.
- Refit k_{bio}, the shear exponent α and the biofouling index B; check the predicted 1.6 months Gulf cleaning interval against reality.

Cost. Analyst time only. **Risk.** Data access; mitigated by accepting anonymised data and offering a confidentiality instrument. **Exit criterion.** Model reproduces the plant's observed cleaning interval and membrane life within a stated tolerance.

Stage 1 — Measure cycle tolerance (3–12 months, parallel)

Objective. Measure N_{tol}, the parameter the whole result turns on.

- Bench rig, one standard seawater element, Gulf feedwater (45 g/L, 30 °C to 35 °C, representative biological load).
- Two parallel regimes: continuous, and daily start/stop with a realistic shutdown duration. Identical cleaning protocol on both.
- Record permeability *not* recovered by cleaning against cumulative cycle count. The slope gives N_{tol} directly.
- Extend with a reduced-load arm (identifies α) and an idle arm (identifies the stagnation multiplier). Our results show all three are needed: removing restart damage alone is not sufficient.

Cost. One bench rig, a few hundred cycles; under a year at daily cycling, less under accelerated protocols. **Natural host.** Khalifa University membrane laboratories, with operator-supplied feedwater. **Exit criterion.** A measured N_{tol} with uncertainty bounds, placing a real membrane on the frontier.

Stage 2 — Shadow-mode decision support (12–18 months)

Objective. Run the model alongside operations without touching control.

- Ingest live feed temperature, salinity and normalised permeability.
- Produce a weekly report: predicted cleaning date, projected membrane life, and the shadow price of flexibility at current conditions.
- Compare predictions against outcomes; this is the credibility-building phase and it carries no operational risk.

Exit criterion. Prediction error inside tolerance over two cleaning cycles.

Stage 3 — Advisory dispatch (18–30 months)

Objective. Put the degradation shadow price in front of the people who schedule the plant.

- Day-ahead advisory: given tomorrow's forecast solar and tariff, the recommended load profile and its membrane-life cost.
- Operators retain full control; the tool advises.
- Where the measured N_{tol} turns out to be high, this is where the benefit is captured. Where it is low, the tool's main value is documenting why flexing is declined — which is itself commercially useful when negotiating offtake terms.

Stage 4 — Design-stage integration (24–36 months)

Objective. Move the insight upstream, where the money is.

- Use the design-space map at tender stage to size PV, RO oversize and product storage jointly, with membrane life inside the objective.
- Feed the carbon-price schedule into concession structuring so that renewable obligations are costed correctly at the bid stage.

Commercial logic for the operator

The immediate value is not efficiency but **risk avoidance**. Our reference case shows a degradation-blind flexible dispatch costing 0.442 USD/m^3 more than the degradation-aware alternative, and a blind model being over-optimistic about its own chosen design by 12.6%. On a plant the scale of Al Taweelah, an error of that proportion on the water tariff is a material commercial exposure over a 25 yr concession. The cost of avoiding it is one bench campaign and an analyst.

What would make us wrong

- A measured N_{tol} above roughly 4201 cycles would make flexible operation defensible, and a large part of the international demand-response literature would then apply to the Gulf directly. Our frontier states exactly what it would then be worth.
- Membrane chemistry substantially more cycling-tolerant than current polyamide would move the same threshold.
- A restart protocol that suppresses the irreversible increment — permeate flushing before shutdown is the obvious candidate — could shift the frontier without any change in membrane material.

Each of these is testable, and each would be a good outcome. The purpose of this roadmap is to find out which one is true before capital is committed on the assumption that flexibility is free.

Contact: Muhammet Yalkapov (team leader), 1095390@students.adu.ac.ae, Abu Dhabi University. Faculty advisor and mentor: Dr. Murad Al-Rajab, Abu Dhabi University. The partner operator and the Stage 1 bench-test laboratory are still to be confirmed.